Q13 Discussion Questions

1. Geva et al. showed FFN layers *can be viewed* as key–value memories, and ROME's causal tracing supported mid-MLP storage for crisp entity facts. Where do you expect this picture to break — procedural knowledge, commonsense, skills — and what experiment would distinguish "stored in MLPs" from "emergent from the whole network"?
2. Superposition trades capacity against interference. If a frontier model stores many more features than dimensions, what does that imply about the achievable precision of knowledge editing — and is there any representation scheme that would give us both compression and editability?
3. Probing accuracy shows information is *accessible* in a representation, not that the model *uses* it. Design a minimal experiment that separates the two for a factual claim (e.g., "the capital of Australia"), using only black-box queries and linear probes.
4. ROME/MEMIT edits can bleed over to neighboring subjects and degrade fluency. Given the superposition story, is bleedover a bug that better engineering will remove, or an intrinsic property of distributed storage? What would count as evidence for each?
5. Knowledge-in-weights is static and hard to update; knowledge-in-context is instant but tiny. Which failure mode worries you more for a deployed assistant — stale parametric knowledge overriding correct context, or context overflow dropping crucial facts — and how would you architect the boundary between the two channels?